Bessel's Differential Equation and Bessel Functions

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The Gamma Function: Motivation

- We first met factorials $n!$ for non-negative integers n. Is there a natural way to make sense of "$x!$" for *any* real x?
- Euler's answer is the **Gamma function**, which interpolates the factorial smoothly between integers.
- It turns out to be indispensable in the study of Bessel's equation: writing the series solution for *general* (non-integer) order requires exactly this function.
- This is precisely the order of development in Simmons: Gamma function first, then Bessel's equation and its solutions.



Definition and Convergence

- For $x > 0$, define

$$\Gamma(x) = \int_0^\infty e^{-t}\, t^{x-1}\, dt.$$

- Convergence: near $t = 0$, t^{x-1} is integrable provided $x > 0$; near $t = \infty$, e^{-t} decays faster than any power of t grows, so the integral converges for every $x > 0$.
- This defines $\Gamma(x)$ as a smooth, positive function on $(0, \infty)$.
- Simmons develops this as the prototype of an "improper integral depending on a parameter" — a recurring theme in the study of special functions.



The Functional Equation

- Integrating by parts,

$$\Gamma(x+1) = \int_0^\infty e^{-t} t^x \, dt = \left[-e^{-t} t^x\right]_0^\infty + x \int_0^\infty e^{-t} t^{x-1} \, dt = x\,\Gamma(x).$$

- This is the fundamental **functional equation**:

$$\boxed{\Gamma(x+1) = x\,\Gamma(x)}$$

- Direct computation: $\Gamma(1) = \int_0^\infty e^{-t} \, dt = 1$.
- Combining these: $\Gamma(2) = 1 \cdot \Gamma(1) = 1$, $\Gamma(3) = 2\Gamma(2) = 2$, and by induction

$$\Gamma(n+1) = n!, \qquad n = 0, 1, 2, \ldots$$

- So $\Gamma(x)$ genuinely extends the factorial function to all positive reals.



A Key Special Value: $\Gamma(1/2)$

- Setting $t = u^2$ in the definition:

$$\Gamma\left(\tfrac{1}{2}\right) = \int_0^\infty e^{-t} t^{-1/2}\, dt = 2\int_0^\infty e^{-u^2}\, du.$$

- The classical Gaussian integral gives $\displaystyle\int_0^\infty e^{-u^2}\, du = \frac{\sqrt{\pi}}{2}$, so

$$\boxed{\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}}$$

- Combined with the functional equation, this generates every half-integer value:

$$\Gamma\left(\tfrac{3}{2}\right) = \tfrac{1}{2}\sqrt{\pi}, \qquad \Gamma\left(\tfrac{5}{2}\right) = \tfrac{3}{2}\cdot\tfrac{1}{2}\sqrt{\pi} = \tfrac{3}{4}\sqrt{\pi}, \qquad \ldots$$

- These half-integer values will reappear directly in the Bessel functions $J_{\pm 1/2}(x)$.



Extending $\Gamma(x)$ to Negative Values

- The integral definition only converges for $x > 0$. But the functional equation, rearranged as

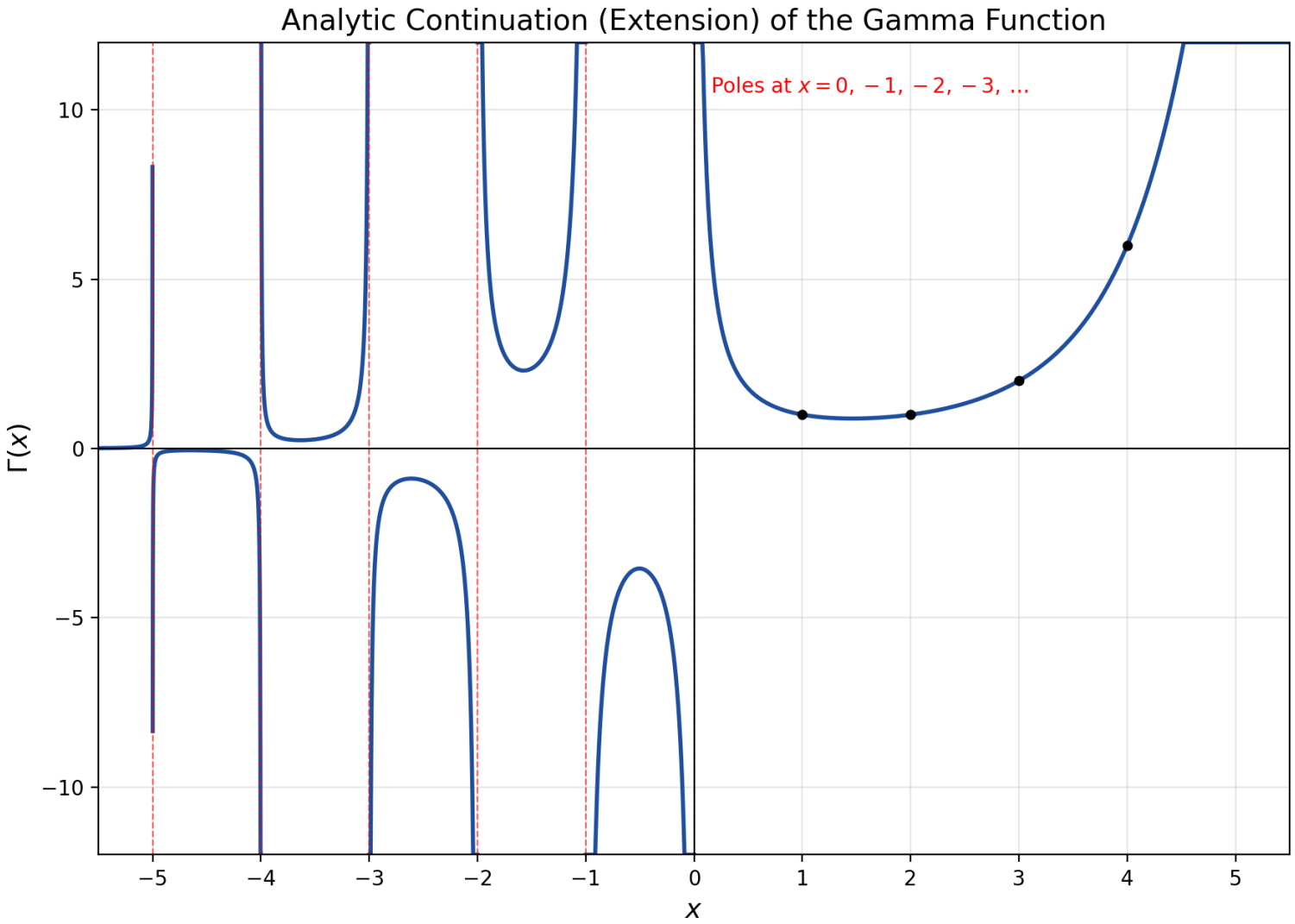
$$\Gamma(x) = \frac{\Gamma(x+1)}{x},$$

lets us **define** $\Gamma(x)$ for $x < 0$ by working backward from positive arguments.

- E.g., for $-1 < x < 0$, define $\Gamma(x) := \Gamma(x+1)/x$, where $x + 1 \in (0, 1)$ is already known.
- Repeating this pushes the definition to $-2 < x < -1$, then $-3 < x < -2$, and so on, covering every negative non-integer.
- Example: $\Gamma\left(-\frac{1}{2}\right) = \dfrac{\Gamma(1/2)}{-1/2} = -2\sqrt{\pi}$.
- At $x = 0, -1, -2, \ldots$ the extension **fails**: $\Gamma(x)$ has a simple pole at every non-positive integer, since dividing by $x = 0, -1, -2, \ldots$ at some stage in the recursion blows up.



Graph of Extended Gamma Function



The Graph of $\Gamma(x)$

- $\Gamma(x)$ is smooth and positive for $x > 0$, with a minimum near $x \approx 1.46$ (where $\Gamma(x) \approx 0.886$), and $\Gamma(x) \to \infty$ as $x \to 0^+$ or $x \to \infty$.
- For $x < 0$, the extended function alternates in sign between consecutive poles: positive on $(-1, 0)$ becomes negative on... in fact it is negative on $(-1, 0)$, positive on $(-2, -1)$, negative on $(-3, -2)$, and so on, alternating sign in each unit interval.
- Vertical asymptotes occur at every $x = 0, -1, -2, -3, \ldots$
- This qualitative picture is essential background before using Γ in the Bessel series, where terms like $\Gamma(n + m + 1)$ appear in every denominator.



Problem: Computing $\Gamma(-5/2)$

- Using the extension $\Gamma(x) = \Gamma(x+1)/x$, compute $\Gamma(-5/2)$.
- Step down from $\Gamma(1/2) = \sqrt{\pi}$:

$$\Gamma\left(-\tfrac{1}{2}\right) = \frac{\Gamma(1/2)}{-1/2} = -2\sqrt{\pi}.$$

$$\Gamma\left(-\tfrac{3}{2}\right) = \frac{\Gamma(-1/2)}{-3/2} = \frac{-2\sqrt{\pi}}{-3/2} = \frac{4}{3}\sqrt{\pi}.$$

$$\Gamma\left(-\tfrac{5}{2}\right) = \frac{\Gamma(-3/2)}{-5/2} = \frac{\frac{4}{3}\sqrt{\pi}}{-5/2} = -\frac{8}{15}\sqrt{\pi} \approx -0.9453.$$



Origins of Bessel's Equation

- **Bessel's equation of order** ν $(\nu \geq 0)$ is

$$\boxed{x^2y'' + xy' + (x^2 - \nu^2)y = 0} \tag{1}$$

- It arises constantly in problems with **circular or cylindrical symmetry**: vibrations of a circular membrane (drum), heat conduction in a cylinder, and the radial part of Laplace's equation in cylindrical or spherical coordinates.
- Historically, Bessel encountered it while studying planetary motion (expansion of Kepler's equation); Euler and Daniel Bernoulli had met special cases earlier.
- Like Legendre's equation, $x = 0$ is a singular point — but here it is (for most ν) not an ordinary point, so we need the full machinery of the Frobenius method.



The Frobenius Setup at $x = 0$

- In standard form, $y'' + \dfrac{1}{x}y' + \dfrac{x^2 - \nu^2}{x^2}y = 0.$
- Since $xP(x) = 1$ and $x^2Q(x) = x^2 - \nu^2$ are both analytic at $x = 0$, the point $x = 0$ is a **regular singular point**.
- To find a series solution, we substitute

$$y = x^r \sum_{n=0}^{\infty} a_n x^n, \qquad x > 0,\ a_0 \neq 0.$$

  into (1).
- Let $\rho(r) = r^2 - \nu^2$. Further computation gives

$$\sum_{n=0}^{\infty} \rho(n+r)a_n x^n + \sum_{n=2}^{\infty} a_{n-2}x^n = 0.$$

- This equation is rearranged as

$$\rho(r)a_0 + \rho(r+1)a_1 x + \sum_{n=2}^{\infty} \Big(\rho(n+r)a_n + a_{n-2}\Big)x^n = 0.$$



The Recurrence Relation

- By the assumption of Frobenius' method ($a_0 \neq 0$ for the indicial equation), we have

$$\rho(r) = 0, \qquad \rho(r+1)a_1 = 0, \qquad \rho(r+n)a_n = -a_{n-2}, \quad n \geq 2.$$

- The **indicial equation** is $r^2 - \nu^2 = 0$.
- Hence $r_1 = \nu$, $r_2 = -\nu$, and $r_1 - r_2 = 2\nu$.
- A Frobenius series solution always exists for the larger root $r = r_1 = \nu$.
- With $r = \nu$, we obtain

$$a_1 = 0, \qquad a_n = -\frac{a_{n-2}}{n(n+2\nu)}, \quad n \geq 2.$$

- Iterating these relations, we have

$$a_{2n+1} = 0, \; a_{2n} = \frac{(-1)^n a_0}{2^{2n} n!(\nu+1)(\nu+2)\cdots(\nu+n)} \quad n \geq 1.$$



Bessel's Eq. (ctn.): The Bessel Function $J_\nu(x)$

- Therefore, the Frobenius series solution is

$$y_1(x) = a_0 x^\nu \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} n!(\nu+1)(\nu+2)\cdots(\nu+n)}\right). \tag{2}$$

- Choosing $a_0 = \dfrac{1}{2^\nu \Gamma(\nu+1)}$ and using $\Gamma(\nu+1)(\nu+1)(\nu+2)\cdots(\nu+n) = \Gamma(n+\nu+1)$, the series (2) becomes the **Bessel function of order ν of the first kind**:

$$\boxed{J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n+\nu+1)} \left(\frac{x}{2}\right)^{2n+\nu}} \tag{3}$$

- For $\nu = 0$: $J_0(x) = 1 - \dfrac{x^2}{4} + \dfrac{x^4}{64} - \cdots$, and $J_0(0) = 1$.
- For $\nu \geq 1$ (or any $\nu > 0$): $J_\nu(0) = 0$, since the lowest power of x present is x^ν.

